F70LB Life Insurance Mathematics B

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11 Income Protection Insurance and Duration Dependence

11.1 Introduction

In this section, we look in more detail at features of disability insurance and long-term care insurance. We will find that this introduces duration dependence of two forms:

(1) Some of the policy **cashflows** may depend on the duration spent in a state. This can happen even if the underlying life-history model is **Markov**.

(2) Some of the **transition intensities** may depend on the length of time (duration) spent in a state. In this case the **Markov** assumption does not hold.

11.2 Features of disability insurance

Disability insurance is commonly called Income Protection Insurance (IPI) in the UK. The benefit it pays is income during periods of incapacity due to sickness and disability.

In the last section we just assumed that premiums were payable continuously while able, benefits were payable continuously while sick, and death benefits might be paid as well. In practice, IPI policies are more complicated.

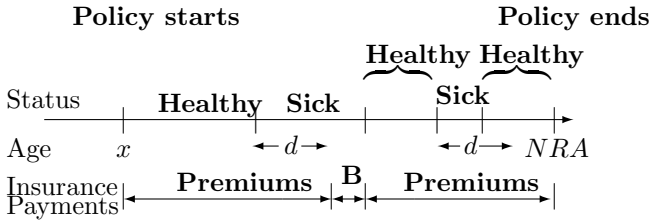
- **Term**: The policies are typically written up to the normal retirement age (NRA).
- **Definition of sickness**: The policy will specify the definition of sickness. A typical definition is:

  > **"totally unable, through sickness or accident, to follow own occupation and not following any other for profit or reward."**

  Note that a definition like "unable to follow any occupation" may be **very restrictive to the policyholder**.
- **Exclusions**: There may be exclusions to inability to work due to pregnancy, drug-related illnesses, etc.
- **Premiums**: The premiums are typically level monthly and payable while benefit is not being received. Premium rates are usually guaranteed at outset.
- **Benefits**: Benefits are usually in the form of a monthly (or weekly) income. This may be fixed, or increase with inflation, or it may fall to a lower level (e.g. half) after some fixed duration of payment, to encourage the policyholder to return to work.
- **Waiver of Premiums**: Since premiums are not paid while the benefit is received, this is also a benefit of waiver of premiums.
- **Deferred Period**: Benefits do not, in fact, start as soon as the policyholder falls sick. They are deferred for a period of time, chosen by the policyholder, during which the policyholder must remain sick, or benefits will not commence. This is called the deferred period. We denote it d (in years). Typical values are 1 week, 4 weeks, 13 weeks, 26 weeks or 52 weeks, i.e. $d \approx 1/52, 1/12, 1/4, 1/2$ or 1 year.

- **Waiting period**: A period of time after taking out a IPI policy, during which the new policyholder is not entitled to sickness benefits. This may be 6 months. It is not the same as the deferred period.
- **Off-Period**: If the benefits are cut after some duration of continuous sickness (see above), the off-period defines the minimum period of good health that must pass before two episodes of sickness can be considered as different.
- **Benefit Limits**: The benefit will be reduced if the policyholder's total net income is greater than, say, 75% of net income before sickness. This removes the chance that a policyholder will be better off sick than at work.
- **Underwriting**: Medical underwriting is similar to, but not the same as, life insurance. In particular, more details about occupation are considered for IPI underwriting than for life insurance underwriting.

The figure below gives an overview of how IPI policies work.



11.3 Duration dependence caused by the deferred period

11.3.1 Occupancy probabilities

By definition, the deferred period means that the payment of sickness benefit depends on the duration of sickness. This is true even if the underlying life history model is Markov. Therefore we define:

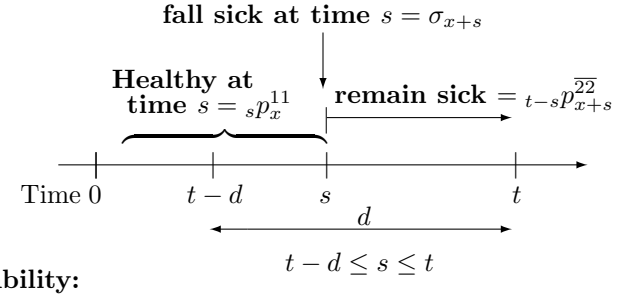
$$
\begin{aligned}
{}_{d,t}p_x^{12} \quad = \quad & \mathrm{P}[\ \textbf{life is sick at age } x+t \textbf{ with duration} \\
& \textbf{of sickness} \leq d \mid \textbf{life is healthy at age } x\ ]
\end{aligned}
$$

Since only the policy cashflows while sick are affected, we do not need to define similar probabilities for any other states.
To find an expression for this in terms of more basic quantities, we consider the *last time at which the policyholder fell sick.* This could have been any time between time $t-d$ and t. There are two cases:

Case 1: $d \geq t$. Since we knew anyway that the life last fell sick between time 0 and time t, knowing that the duration of sickness is $\leq d$ gives us no additional information, so:

$$
{}_{d,t}p_x^{12} = {}_tp_x^{12}.
$$

Case 2: $d < t$. The event (x) fell sick for the *last* time at time s requires (x) to be healthy at age $x+s$, to fall sick before age $x+s+ds$, and then to remain sick for duration $t-s$. See the following diagram:



This event has probability:

$$ {}_sp_x^{11}\,\sigma_{x+s}\,ds\,{}_{t-s}p_{x+s}^{\overline{22}} $$

and since s lies between $t-d$ and t:

$$ {}_{d,t}p_x^{12} = \int_{s=t-d}^{t} {}_sp_x^{11}\,\sigma_{x+s}\,{}_{t-s}p_{x+s}^{\overline{22}}\,ds. $$

This can be evaluated by numerical integration.

11.3.2 EPVs for duration-dependent cashflows

When benefits are duration-dependent, Thiele's equations do not hold. However, it is still possible to calculate premiums and reserves in various ways, although we lose the extreme generality and flexibility that Thiele's equations give us.
We show here one way to calculate IPI premiums when the policy has a deferred period. It is by analogy with the EPV of a life annuity, payable continuously, which is:

$$ EPV = \bar{a}_x = \int_0^{\infty} v^t\,{}_tp_x\,dt. $$

This has the following interpretation: the annuity dt is payable at time t if the status of being alive is fulfilled. This has probability ${}_tp_x$ and payment at time t is discounted by v^t.
Now apply the same reasoning to the following IPI policy issued to a healthy life aged x.

- The policy term is n years.
- Premiums at rate $\bar{P}$ per annum are payable continuously while the life is healthy **or** the life is sick with a duration less than or equal to d years.
- sickness benefit is payable continuously at rate $\bar{B}$ per annum while the life is sick with duration greater than d years.

Premiums are payable at time t as long as the status 'alive and not sick for longer than d years' is fulfilled, which has probability

$$ \left({}_tp_x^{11} + {}_{d,t}p_x^{12}\right) $$

Benefits are payable at time t as long as the status 'alive and sick for longer than d years' is fulfilled, which has probability

$$ \left({}_tp_x^{12} - {}_{d,t}p_x^{12}\right) $$

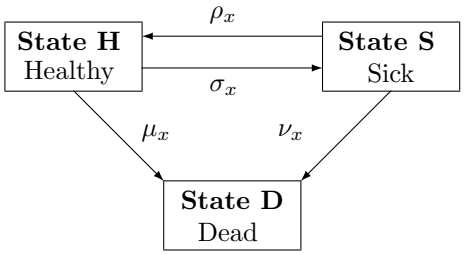
Summing (integrating) the discounted cashflows over the policy term we have:

$$\bar{P} \int_{t=0}^{n} v^t \left({}_tp_x^{11} + {}_{d,t}p_x^{12} \right) dt = \bar{B} \int_{t=0}^{n} v^t \left({}_tp_x^{12} - {}_{d,t}p_x^{12} \right) dt$$

from which $\bar{P}$ can be found. A similar approach can be used in many cases.

Example:

The following 3 state model is used to price various sickness policies. The forces of transition σ, ρ, μ and ν depend only on age.



The following probabilities are defined:

${}_tp_x^{ij}$ is the probability that a life aged x in state i will be in state j at age $x+t$;

${}_tp_x^{\overline{ii}}$ is the probability that a life age x in state i will remain in state i until age $x+t$;

Using these probabilities and/or forces of transition, write down an expression for the expected present value of each of the following sickness benefits for a life currently aged 35 and healthy. The constant force of interest is δ.

1. £1,000 per annum payable continuously while sick, but all benefits cease at age 65
2. £1,000 per annum payable continuously while in the sick state for any continuous period in excess of a year. However, any benefit period is limited to 5 years payments, but the number of possible benefit periods is unlimited
3. £1,000 per annum payable continuously throughout the first period of sickness only

Solution:

(1)

$$\text{EPV} = 1,000 \int_0^{30} \mathbf{e}^{-\delta\, t}\, {}_tp_{35}^{HS}\, dt$$

(2)

$$\text{EPV} = 1,000 \int_0^{\infty} \mathbf{e}^{-\delta\, t}\, {}_tp_{35}^{HH} \sigma_{35+t} \int_1^6 \mathbf{e}^{-\delta\, r}\, {}_rp_{35+t}^{\overline{SS}}\, dr\, dt$$

(3)

$$\text{EPV} \quad = 1,000 \int_0^\infty \mathbf{e}^{-\delta\, t}\, {}_tp_{35}^{\overline{HH}}\, \sigma_{35+t} \int_0^\infty \mathbf{e}^{-\delta\, r}\, {}_rp_{35+t}^{\overline{SS}}\, dr\, dt$$

11.4 Evidence for duration-dependent IPI intensities

There is strong evidence that the intensities of transitions out of the ill state depend on how long a life has been ill as well as on age. This is illustrated in the model in Figure 1, where:

x	represents age at policy inception
$x+t$	represents current age
z	represents current duration of sickness.

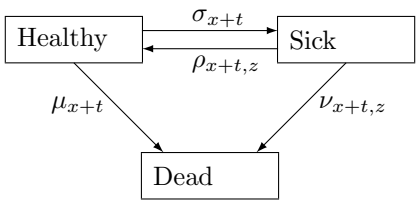


Figure 1: A semi-Markov IPI model.

Note that $\rho_{x+t,z}$ is interpreted as:

$$\rho_{x+t,z}\,dt \;\approx\; \mathrm{P}[\ \textbf{life age } x+t\textbf{, sick for duration } z \textbf{ will recover before age } x+t+dt\ ]$$

(and similarly for $\nu_{x+t,z}$).
A multiple-state model in which any intensity depends on the duration of stay in a state is called a semi-Markov model.

11.5 EPVs where transition intensities are duration-dependent

The same approach as we used to deal with deferred periods can be extended to the semi-Markov model. Consider, for example, the following policy issued to a life aged x:

- premiums ceasing at age 65
- premiums at rate $\bar{P}$ per annum payable continuously while healthy and waived while the benefit is payable
- benefits at rate $\bar{B}$ per annum payable continuously while sick with duration $\geq d$
- interest at i per annum effective
- no expenses.

The equation of value is:

$$\bar{P}\int_{t=0}^{65-x} v^t \left({}_tp_x^{11} + {}_{d,t}p_x^{12}\right) dt = \bar{B}\int_{t=0}^{65-x} v^t \left({}_tp_x^{12} - {}_{d,t}p_x^{12}\right) dt$$

We now suppose that the annual rate of sickness benefit is:

$$\begin{aligned} 0 \quad &\text{for} \quad \textbf{duration} \ \leq d_1 \\ \bar{B}_1 \quad &\text{for} \quad d_1 < \textbf{duration} \ \leq d_1 + d_2 \\ \bar{B}_2 \quad &\text{for} \quad \textbf{duration} \ \geq d_1 + d_2 \end{aligned}$$

with premiums waived while any benefit is payable. The equation of value is then:

$$\begin{aligned} \bar{P} \int_{t=0}^{65-x} v^t \left({}_tp_x^{11} + {}_{d_1,t}p_x^{12} \right) dt \quad &= \quad \bar{B}_1 \int_{t=0}^{65-x} v^t \left({}_{d_1+d_2,t}p_x^{12} - {}_{d_1,t}p_x^{12} \right) dt \\ &+ \quad \bar{B}_2 \int_{t=0}^{65-x} v^t \left({}_{t,t}p_x^{12} - {}_{d_1+d_2,t}p_x^{12} \right) dt. \end{aligned}$$

12 Long-term care insurance

12.1 Background

Long-term care is care provided to those people who are no longer able to look after themselves, typically the elderly. LTC can:

- range from 2 hours a week to 24 hours a day
- be informal, from a spouse or children (usually in the home)
- be formal, on a paid-for basis, typically provided in a nursing home or residential home.

Long-term care (LTC) insurance is a product designed to fund, or partially fund, long-term care.

In the UK, at the time of writing, there is not a large market in long-term care insurance products, even though in principle it is a risk that is suited to an insurance solution.

The most common LTC insurance product sold in the UK is called an immediate needs annuity. It is normally bought after a need for care arises.

12.2 Product Design

We describe some possible LTC insurance designs here:

- Stand-alone: Premiums can be single lump-sum or regular until time of claim or death. Benefits are in the form of income for the duration of a valid claim.
- LTC as a rider to a whole life plan: In the event of satisfying the claims criteria the death benefit is accelerated and payable in monthly installments.
- LTC as an extension to IPI: Before NRA, IPI claims criteria and benefit level are used. After this age, LTC claims criteria used are with the same or increased level of benefits. Premiums may remain at the same level, decrease, or stop at the change-over age.
- Immediate needs annuities (mentioned above): These are effectively impaired life annuities with the aim of covering the costs of current care or care that will be required in the near future. A single lump-sum premium provides a guaranteed monthly annuity to cover part or all of care costs for as long as care is required.

12.3 Product Features

Claims Criteria

Claims can usually be triggered through physical disability or cognitive impairment.

Typical claims criteria would be:

- failing 2 or 3 out of a benchmark of 6 activities of daily living (ADLs); or
- failing a cognitive impairment test.

The Association of British Insurer's (ABI's) benchmark ADLs are: washing, dressing, feeding, toileting, mobility and transferring.

LTC insurance products may pay:

- benefits towards care costs up to a maximum sum assured; or
- a fixed percentage of the sum assured on triggering of benefits. For example:
  50% of SA on the failure of 2 ADLs
  100% of SA on the failure of 3 or more ADLs.

Failure of 3 ADLs is typically used as the point for payment of the full sum assured, since this is the point when residential care will usually be required.

Benefit Limits

Limits may be imposed on:

- the length of the benefit period (eg 3 or 5 years)
- the total amount of benefits payable.

Benefit Escalation

Benefits may be level or indexed. Types of indexation:

- fixed % per annum (eg 5%)
- linked to an index, usually with a cap (eg RPI up to a maximum of 15% p.a.).

Premiums

Premiums can be regular or single lump-sum.

Regular premium policies are generally reviewable (i.e. the insurance company can increase them at their discretion).

Guarantees are sometimes offered, for example, the premium rates may be guaranteed to remain level for 10 years, but be reviewable annually thereafter.

It is normal for regular premiums to escalate in line with the benefits.

Premium Waiver

Waiver of premium applies to all regular premium contracts, such that premiums are not payable while benefits are being paid.

Deferred Period

This is the period of time a claimant must continuously fail the claims criteria before benefits are paid.

Typically this is 3 months, although there are also deferred periods of 6, 12, 24 and 36 months.

Other features

Since LTC is a protection product there is not normally any surrender value or death benefit.

12.4 Pricing and reserving for LTC products

To find premiums and policy values for a LTC insurance contract, we define a suitable multiple-state model of the insured person's life history. We have already covered Markov multiple-state models but we should always remain aware that the data may suggest that a semi-Markov model is more appropriate.

Here, we give another example to show that the approach can deal with the complexity of LTC contracts. Consider the following LTC policy, with premiums payable continuously at annual rate $\bar{P}$ and:

- SA of $£B$ pa
- 50% of SA payable on loss of 2 ADLs
- 100% of SA payable on loss of 3 or more ADLs.

We represent the life history underlying this contract using the following model.
We can calculate the premium rate $\bar{P}$ from Thiele's differential equations:

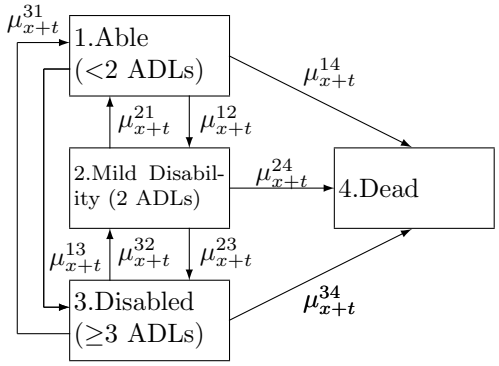


Figure 2: A Markov model of long-term care insurance, allowing for two levels of disability.

$$
\begin{aligned}
\frac{d}{dt}V^1(t) &= \delta\, V^1(t) + \bar{P} - \mu^{12}_{x+t}\,(V^2(t) - V^1(t)) \\
&\quad -\mu^{13}_{x+t}\,(V^3(t) - V^1(t)) + \mu^{14}_{x+t}\,V^1(t) \\
\frac{d}{dt}V^2(t) &= \delta\, V^2(t) - \frac{B}{2} - \mu^{21}_{x+t}\,(V^1(t) - V^2(t)) \\
&\quad -\mu^{23}_{x+t}\,(V^3(t) - V^2(t)) + \mu^{24}_{x+t}\,V^2(t) \\
\frac{d}{dt}V^3(t) &= \delta\, V^3(t) - B - \mu^{31}_{x+t}\,(V^1(t) - V^3(t)) \\
&\quad -\mu^{32}_{x+t}\,(V^2(t) - V^3(t)) + \mu^{34}_{x+t}\,V^3(t) \\
\frac{d}{dt}V^4(t) &= 0
\end{aligned}
$$

using the boundary conditions:

$$V^i(\omega - x) = 0 \quad \textbf{for} \quad i = 1, 2, 3, 4$$

where ω is the oldest age in the life table.